Nonequilibrium Dynamics of the Two-Dimensional Ising Model Under Periodic Magnetic Driving

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Abstract

In this study, we investigate the nonequilibrium behavior of the two-dimensional Ising model under a periodically oscillating external magnetic field using Metropolis Monte Carlo simulations. We focus on how the magnetization dynamics and hysteresis response depend on the driving period, temperature, and lattice size.

We observe a non-monotonic dependence of the hysteresis response on the driving period, with the largest response occurring at an intermediate period. As the temperature decreases, the period corresponding to the maximum response shifts toward larger values, consistent with a slower magnetization reversal process. By introducing the dynamic order parameter, we also distinguish persistent magnetization bias under fast driving from the more symmetric magnetization reversal observed at longer periods.

Our finite-size investigation suggests that the hysteresis response is relatively insensitive to lattice size within the investigated range, while the dynamic order parameter exhibits stronger size dependence. Numerical validation supports the simulation implementation and several qualitative findings, although limitations in equilibration and periodic steady-state sampling remain.

These results illustrate how thermal fluctuations, driving timescales, and finite-size effects influence the nonequilibrium dynamics of the Ising model.

Project Background:

This project builds upon the Physics 77 Capstone Project, in which our group investigated the two-dimensional Ising model. My work focused on Group A's computational investigation of magnetization dynamics under a periodically oscillating external magnetic field, particularly how the magnetization responds to changes in the driving period.

Building on this initial investigation, the present study aims to explore a broader question: How do the magnetization dynamics and hysteretic response of the system change under different physical conditions? Specifically, we examine the effects of the driving period, temperature (T), and lattice size (L), and investigate whether the trends observed under a particular set of conditions remain consistent across a wider range of parameters.

An additional objective is to assess the finite-size dependence of the numerical results and determine to what extent a finite two-dimensional Ising lattice can approximate the behavior of the system in the thermodynamic limit.

Model and Simulation Method

We start by considering a single spin, which can take one of two possible states: s = +1 or s = -1.

We then construct a two-dimensional square lattice with side length L, containing a total of N = L^2 spins. Each spin interacts with its nearest neighbors and with an external magnetic field.

The total energy of the system is described by the Ising Hamiltonian:

$$E = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i$$

Here, the first summation runs over all nearest-neighbor pairs, counted once. The constant J represents the interaction strength between neighboring spins, while h represents the strength of the external magnetic field. We assume $J > 0$, so neighboring spins energetically favor alignment.

We use periodic boundary conditions, meaning that spins along one edge of the lattice are connected to spins on the opposite edge. As a result, every spin has four nearest neighbors, regardless of its position.

To investigate how the system behaves at a given temperature T, we use the Metropolis Monte Carlo algorithm [1]. The idea is to generate a sequence of spin configurations through random updates, with their acceptance probabilities determined by the corresponding energy changes.

For each attempted update, we randomly select one spin and propose flipping its orientation from s_i to $-s_i$. We then calculate the corresponding energy difference, ΔE.

If $\Delta E \leq 0$, the proposed flip is always accepted. Otherwise, it is accepted with probability exp($-\Delta E$/T), where we use reduced units with k_B = 1.

One Monte Carlo sweep (MCS) consists of N = L^2 attempted spin flips. This serves as the unit of simulation time, although it does not correspond to a fixed amount of physical time.

For a fixed temperature and external field, this procedure allows the system to sample configurations consistent with the Boltzmann distribution after equilibration.

In our study, however, the external magnetic field varies periodically with time. We therefore use the same local update rule to investigate how the magnetization responds to an oscillating field, potentially driving the system away from equilibrium.

Now, for each attempted spin flip, we consider the change in the total energy of the system. Let H_i be the energy of the configuration before the proposed flip, and H_f be the energy after the flip. The energy difference is given by:

$$\Delta E = H_f - H_i$$

Since only one spin is flipped, all energy terms that do not involve this spin remain unchanged. Therefore, for a spin s_i, the energy difference can be written as:

$$\Delta E = 2s_i \left(J \sum_{j \in \text{nn}(i)} s_j + h \right)$$

where the summation runs over the nearest neighbors of spin i.

We know that, at thermal equilibrium, the probability of a given configuration is proportional to its Boltzmann weight:

$$P(A) = \frac{\exp[-H(A)/T]}{Z}$$

where A represents a configuration, Z is the partition function, and we use k_B = 1.

Suppose we propose a spin flip that changes the system from configuration A to configuration B. Their energies are $H(A)$ and $H(B)$, respectively, with ΔE = $H(B) - H(A)$.

The ratio of their equilibrium probabilities is then given by:

$$\frac{P(B)}{P(A)} = \frac{\exp[-H(B)/T]}{\exp[-H(A)/T]}$$

Therefore,

$$\frac{P(B)}{P(A)} = \exp(-\Delta E/T)$$

Notice that the partition function Z cancels out. This means that we can determine the relative Boltzmann probabilities of two configurations using only their energy difference.

Our goal is to construct a random update process such that, after a sufficiently long time, the probability distribution of the system approaches the Boltzmann distribution.

To achieve this, we need to satisfy two requirements.

First, the Boltzmann distribution should be a stationary distribution. This means that once the system reaches equilibrium, the probability of each configuration should remain unchanged after an update, even though individual spin configurations may continue to change.

Mathematically, for any configuration B:

$$P(B) = \sum_A P(A)K(A \to B)$$

where K(A → B) is the probability of transitioning from configuration A to configuration B in one update. The summation includes all possible initial configurations, including B itself.

Second, we want the system to converge to this distribution from different initial configurations. In other words, the Boltzmann distribution should not only be stationary, but should also be the long-time distribution reached by our simulation.

To guarantee the stationary-distribution requirement, we impose a stronger condition called detailed balance:

$$P(A)K(A \to B) = P(B)K(B \to A)$$

This means that, at equilibrium, the probability flow from A to B is equal to the probability flow from B to A.

Now, each transition consists of two steps: proposing a spin flip and deciding whether to accept it.

We can therefore write:

$$K(A \to B) = q(A \to B)a(A \to B)$$

where q(A → B) is the proposal probability, and a(A → B) is the acceptance probability.

Since we randomly select one spin at a time, the proposal probabilities for a given spin flip and its reverse are equal:

$$q(A \to B) = q(B \to A)$$

Substituting this into the detailed balance condition, we obtain:

$$P(A)a(A \to B) = P(B)a(B \to A)$$

Therefore,

$$\frac{a(A \to B)}{a(B \to A)} = \frac{P(B)}{P(A)} = \exp(-\Delta E/T)$$

This condition determines the ratio of the forward and reverse acceptance probabilities, but it does not uniquely determine their individual values.

We therefore choose the Metropolis rule, which provides a simple way to satisfy detailed balance while maximizing the acceptance probability for each proposed transition.

If $\Delta E \leq 0$, the proposed flip lowers or maintains the energy, and we always accept it.

If $\Delta E > 0$, we accept the flip with probability $\exp(-\Delta E/T)$.

Combining these two cases gives:

$$a(A \to B) = \min\{1, \exp(-\Delta E/T)\}$$

By using this acceptance rule, we construct a Markov chain that has the Boltzmann distribution as its stationary distribution. For our finite lattice at T > 0, random single-spin proposals also allow the system to explore the configuration space and converge to this equilibrium distribution.

One Monte Carlo sweep (MCS) is defined as L^2 attempted spin flips. We use MCS as the unit of simulation time.

In our nonequilibrium study, the external magnetic field changes periodically:

$$h(t) = h_0 \sin\left(\frac{2\pi t}{P}\right)$$

At each update, the energy difference is calculated using the current external field. Although the Metropolis rule is consistent with the equilibrium Boltzmann distribution for a fixed field, the periodically driven system does not necessarily remain in instantaneous thermal equilibrium.

This allows us to investigate the magnetization response and hysteresis under periodic driving.

```
Initialize an L x L spin lattice

For each Monte Carlo sweep t:

    Set h(t) = h0 sin(2*pi*t/P)

    Repeat L^2 times:

        Select a random site (i, j)

        Calculate:
            Delta E = 2*s(i,j)*
                (J*sum(neighbor spins) + h(t))

        If Delta E <= 0:
            Flip the selected spin
        Otherwise:
            Flip with probability exp(-Delta E/T)

    Record m(t) = mean(all spins)
```

Figure 1: Pseudocode of the Metropolis simulation for the periodically driven two-dimensional Ising model. Each Monte Carlo sweep consists of L^2 random-site flip attempts, with the external field held constant during the sweep. Periodic boundary conditions are applied when evaluating neighboring spins.

Supplementary visualization. To illustrate how the spin configuration evolves under periodic driving, an animated demonstration is available on GitHub:

View Spin Configuration Dynamics.

Organization of the Research Note

The following section is devoted to numerical validation, where we examine the implementation of our simulation, compare its equilibrium statistics with known theoretical results, and investigate the reliability of our periodically driven simulations.

Since this section focuses mainly on numerical methodology and convergence, readers who are primarily interested in the physical results may proceed directly to **The Driving Period**, where our main investigation begins.

However, the validation results provide important context for understanding the reliability and limitations of the subsequent conclusions.

Numerical Validation

Before investigating the nonequilibrium behavior of the system, we need to verify that our numerical implementation correctly reproduces the expected properties of the Ising model and that our simulation results are sufficiently reliable for further physical interpretation.

There are two main questions we want to address. First, does our code correctly implement the Metropolis algorithm and reproduce the expected equilibrium statistics? Second, when we introduce the periodically oscillating external magnetic field, can we obtain sufficiently stable measurements of the system's response?

These are two different requirements. Even if the Metropolis algorithm is correctly implemented, the system may still require a long time to reach equilibrium or a stable periodic response. Therefore, a correct update rule alone does not guarantee that every numerical result has converged.

We divide our validation into three stages: verification of the numerical implementation, comparison with known equilibrium results, and reliability checks for the periodically driven system.

1. Verification of the Metropolis Implementation

The first thing we want to verify is that the energy difference calculated by our code agrees with the Hamiltonian introduced previously.

For a proposed flip of spin (s_i), the Metropolis algorithm uses

$$\Delta E = 2s_i \left(J \sum_{j \in \mathrm{nn}(i)} s_j + h \right).$$

Since this expression only considers the flipped spin and its nearest neighbors, we can independently calculate the total Hamiltonian before and after the flip and compare the two energy differences:

$$\Delta E_{\mathrm{local}} \stackrel{?}{=} H_{\mathrm{after}} - H_{\mathrm{before}}.$$

We performed 444 such comparisons using the original simulator. All tested energy differences agreed with the independently calculated values.

We also checked whether adding an energy-recording function changes the original simulation. Since recording energy should not affect the spin updates or consume additional random numbers, the magnetization evolution should remain identical for the same initial configuration and random seed.

In 24 seeded trajectory comparisons, the original simulation and the version with energy recording produced identical magnetization trajectories.

These results support the internal consistency of our numerical implementation. However, they do not by themselves verify that the simulated system has reached thermal equilibrium.

2. Equilibrium Validation

To investigate the equilibrium properties of the model, we first remove the external magnetic field by setting $h = 0$.

For a fixed temperature and lattice size, we expect the system to approach the same equilibrium statistics regardless of its initial configuration, provided that sufficient equilibration time is allowed.

However, equilibrium does not mean that the energy or magnetization becomes constant. Individual spin configurations continue to fluctuate, while the equilibrium probability distribution and its statistical averages remain stationary.

2.1. Initial-State Independence and Energy Stability

We first compare simulations starting from two different initial configurations: an all-up configuration and a random configuration.

For each simulation, we record the energy per spin,

$$e(t) = \frac{H(t)}{L^2}.$$

After discarding the initial relaxation period, we compare the average energy obtained from the two initial states:

$$\langle e \rangle_{\text{all-up}} \approx \langle e \rangle_{\text{random}}.$$

We also compare the averages obtained from different measurement windows to check whether the energy is still undergoing systematic changes.

We perform these tests for three lattice sizes, $L = 20, 40, 80$, and four temperatures, $T = 1.2, 2.0, 2.27, 3.2$, giving 12 parameter combinations. Each condition includes eight independent simulations for each initial state.

To evaluate the results, we introduce a diagnostic tolerance of (0.005) in energy per spin. We consider a test supported when its entire approximate 95% confidence interval lies within the specified tolerance.

Eight of the twelve combinations satisfy all applicable energy-stability criteria. The remaining four combinations are inconclusive: $T = 2.27$ for all three lattice sizes, and $T = 1.2$ with $L = 80$.

An interesting observation occurs at $T = 1.2, L = 80$. Some random initial configurations undergo very slow relaxation, and one run remains at a noticeably higher energy even after 80,000 MCS. This indicates that a

nearly constant energy over a particular time interval does not necessarily guarantee sufficient equilibration.

Near $T = 2.27$, the statistical uncertainties also remain too large to establish energy stability within our selected tolerance.

Therefore, these tests suggest that the equilibration behavior depends on both temperature and lattice size. A fixed number of Monte Carlo sweeps may not be sufficient for all conditions.

However, these inconclusive results do not necessarily indicate an error in the Metropolis algorithm. They instead show that the available simulations do not always provide enough evidence of convergence within our chosen precision.

2.2. Comparison with the Exact Boltzmann Distribution

Although initial-state independence provides evidence of equilibration, it does not directly verify that the sampled equilibrium statistics agree with the Boltzmann distribution.

To perform a more direct test, we consider small lattices where all possible spin configurations can be enumerated.

For a lattice of size $L \times L$, there are

$$2^{L^2}$$

possible configurations. Therefore, for $L = 3$ and $L = 4$, the total numbers of configurations are 512 and 65,536, respectively.

For each configuration $\mathcal{C}$, we calculate its energy (H$\mathcal{C}$) independently. The partition function is then given by

$$Z = \sum_{\mathcal{C}} e^{-H(\mathcal{C})/T}.$$

Using the Boltzmann distribution, we can calculate the exact equilibrium expectation of an observable (O):

$$\langle O \rangle_{\text{exact}} = \frac{\sum_{\mathcal{C}} O(\mathcal{C}) e^{-H(\mathcal{C})/T}}{\sum_{\mathcal{C}} e^{-H(\mathcal{C})/T}}.$$

We apply this calculation to the energy per spin and absolute magnetization:

$$\langle e \rangle, \qquad \langle |m| \rangle.$$

We then run the original Metropolis simulation under the same temperature and lattice size, using 16 independent runs per condition, and compare the numerical results with the exact values.

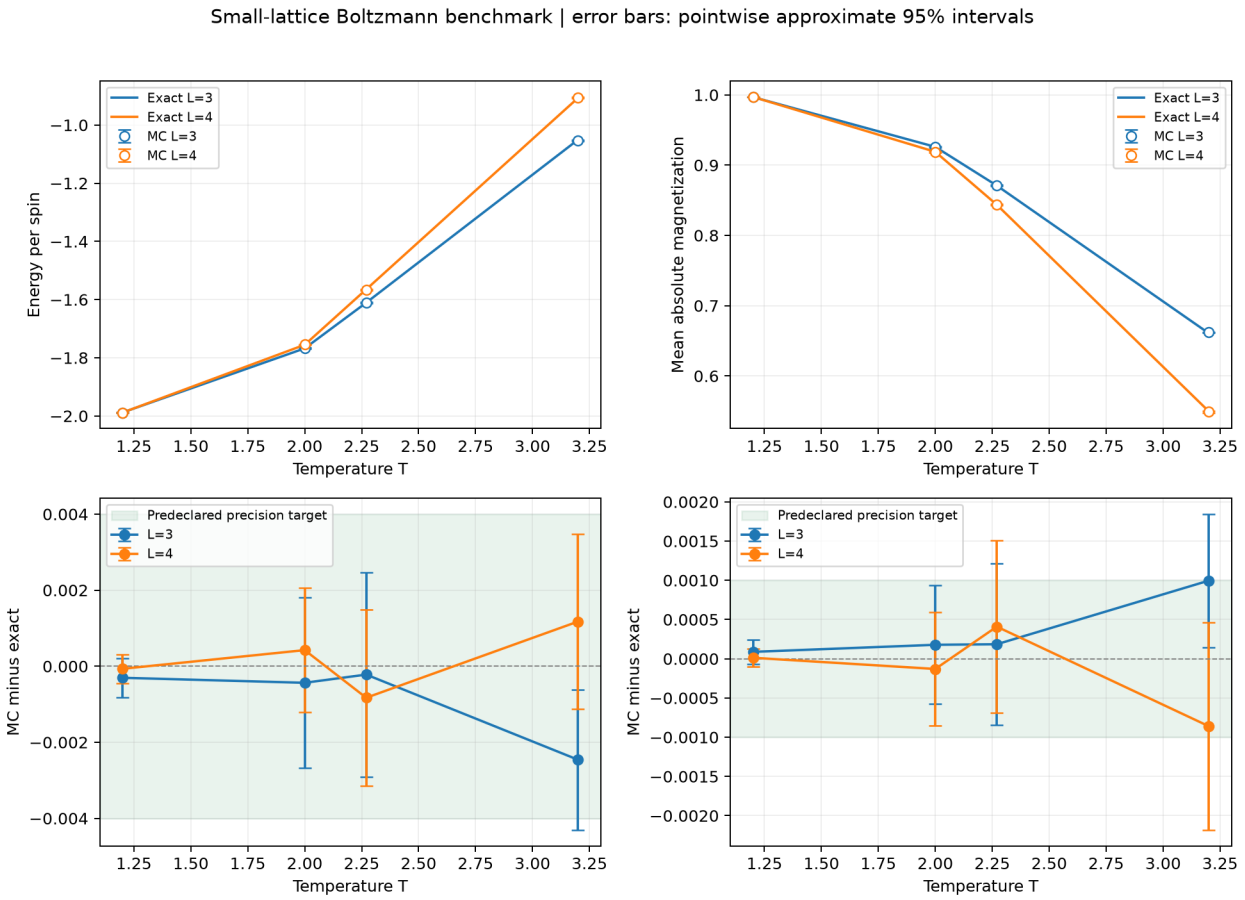


Figure V1. Exact Boltzmann benchmark for $L = 3$ and $L = 4$. The upper panels compare the exact equilibrium energy per spin and absolute magnetization with the corresponding Monte Carlo estimates. The lower panels show the residuals, defined as Monte Carlo minus exact, with approximate pointwise 95% confidence intervals calculated across independent runs. The shaded regions indicate the predefined numerical precision targets: ± 0.004 for energy per spin and ± 0.001 for absolute magnetization.

From Figure V1, we can see that the Monte Carlo results generally follow the exact equilibrium predictions over the investigated temperature range.

However, the residual plots show that visual agreement between the curves is not sufficient to establish agreement within a particular numerical precision.

Among the eight parameter combinations, seven satisfy the selected energy precision criterion, while four satisfy the absolute-magnetization criterion. Four combinations satisfy both direct comparisons. When additional tests of burn-in sensitivity, measurement length, and late-time drift are included, only the two conditions at $T = 1.2$ satisfy every criterion.

In particular, for $L = 3, T = 3.2$, the simulated energy and absolute magnetization show small but statistically distinguishable deviations from the exact values under the reported pointwise intervals. The present tests do not establish a unique explanation for these deviations.

Overall, the comparisons provide evidence that the original Metropolis implementation reproduces important features of the intended Boltzmann statistics. However, some results remain inconclusive at our chosen precision, and additional sampling would be required for stronger quantitative agreement.

2.3. Comparison with the Onsager–Yang Result

We also compare the existing larger-lattice results with the known spontaneous magnetization of the two-dimensional Ising model in the thermodynamic limit.

For the zero-field model, the equilibrium spontaneous magnetization is [3]

$$m_\infty(T) = \begin{cases} \left[1 - \sinh^{-4}(2J/T)\right]^{1/8}, & T < T_c, \\ 0, & T \geq T_c. \end{cases}$$

where $T_c \approx 2.269$ for $J = k_B = 1$ [2].

Our finite-lattice simulations reproduce the expected qualitative temperature dependence: strong magnetization at lower temperatures and weaker magnetization at higher temperatures.

However, we do not expect the finite-system average absolute magnetization to agree exactly with the infinite-system spontaneous magnetization. In particular, finite-size rounding occurs near the critical temperature, and $\langle |m| \rangle$ can remain positive above (T_c) because of finite-size fluctuations.

The comparison is therefore useful as a theoretical reference, but does not provide a rigorous verification of the thermodynamic limit. The full comparison is included in the supplementary validation results.

3. Reliability of the Periodically Driven Simulation

The previous tests support the basic numerical implementation and its equilibrium behavior. However, our main research concerns a system driven by a time-dependent external magnetic field:

$$h(t) = h_0 \sin\left(\frac{2\pi t}{P}\right).$$

In this case, the system does not necessarily remain in instantaneous thermal equilibrium. Therefore, the equilibrium validation cannot by itself guarantee that our nonequilibrium measurements are sufficiently stable.

Instead, we need to determine whether the observables become statistically consistent after the initial transient cycles have been discarded.

We focus on two quantities introduced in the following sections: the cycle-averaged magnetization (Q_k), and the signed work (W_k) performed by the changing external field.

For each complete driving cycle,

$$Q_k = \frac{1}{P} \sum_{j=0}^{P-1} m(kP + j).$$

We retain the signed (Q_k) to examine the evolution of the magnetization direction, while the dynamic order parameter reported in our main experiments is obtained by averaging (|Q_k|) over retained cycles and independent runs.

For the work, we use the discrete expression consistent with the external-field update sequence implemented in the simulator:

$$W_k = -\sum_{j=0}^{P-1} m_{k,j} \left(h_{k,j+1} - h_{k,j} \right),$$

where the field sequence is closed periodically.

This signed work is distinguished from the absolute trapezoidal hysteresis integral used in some of our earlier plots. Both quantities are retained separately in the validation data.

3.1. Periodic-State Stability

We first investigate three representative driving periods, $P = 10, 125, 300$, while keeping $L = 40, T = 2.0$, and ($h_0 = 0.5$) fixed.

For each condition, we compare simulations starting from all-up and random configurations. We also examine the sensitivity of the measured work and dynamic order parameter to the discarded transient cycles, measurement duration, and late-time drift.

The purpose is not to require every configuration to repeat itself after one driving cycle. Instead, we want to examine whether the statistical properties of the cycle-by-cycle observables have become sufficiently stable.

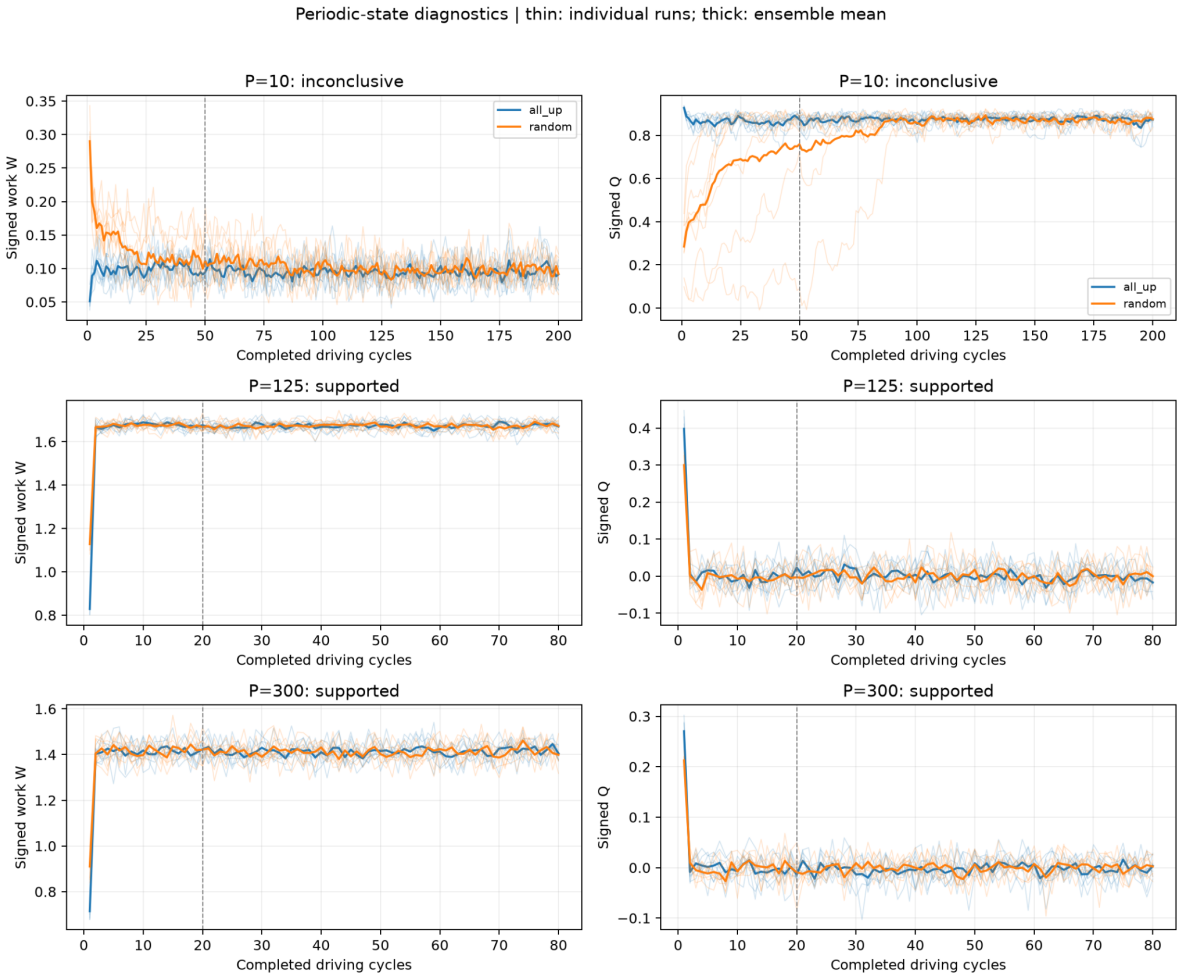


Figure V2. Periodic-state diagnostics for $P = 10, 125, 300$, with $L = 40, T = 2.0$, and $(h_0 = 0.5)$. The left column shows the signed work per cycle, while the right column shows the signed cycle-averaged magnetization. Thin curves represent individual runs, and thicker curves represent the ensemble means for all-up and random initial configurations. Dashed vertical lines indicate the burn-in cutoffs. The labels supported and inconclusive refer to the selected stability tests and numerical tolerances, rather than a mathematical proof of a periodic steady state.

From Figure V2, we can see that the behavior depends significantly on the driving period.

For $P = 125$ and $P = 300$, the cycle-by-cycle observables become relatively stable after the initial transient cycles. The measurements from the two initial-state groups are also consistent within our selected numerical tolerances. These parameter combinations satisfy the additional stability criteria used in the validation.

However, for $P = 10$, the situation is more complicated. The system can maintain a strong magnetization bias over many cycles, but the approach to this behavior depends more strongly on the initial configuration and the early evolution of the system.

Even after extending the simulation to 200 driving cycles, the measurements of $\langle |Q| \rangle$ remain sensitive to the burn-in and measurement windows. Therefore, the $P = 10$ results remain inconclusive under our selected stability criteria.

This does not mean that the observed persistent magnetization is necessarily incorrect. Rather, it means that we cannot yet establish its precise long-time statistical value with the same confidence as the measurements at

$P = 125$ and $P = 300$.

3.2. Checking the Hysteresis Response

We also performed a driving-period scan using the standardized signed-work definition.

For the investigated period grid, the work shows a non-monotonic dependence on the driving period, with the largest measured value occurring at $P = 125$ MCS.

The mean signed work at this period is

$$\langle W(125) \rangle = 1.6739 \pm 0.0029,$$

where the uncertainty is the approximate 95% interval half-width across independent runs.

To examine whether this observed maximum is distinguishable from its neighboring period values, we compare the results using matched seed identities:

$$W(125) - W(100) = 0.04149 \pm 0.00733,$$

$$W(125) - W(150) = 0.02063 \pm 0.00412.$$

Both differences are positive within their reported pointwise intervals, providing additional support for a local maximum near $P = 125$ on the sampled period grid.

The earlier absolute trapezoidal hysteresis integral also reaches its maximum at $P = 125$ in this scan, although its numerical values are not identical to those of the signed work.

Thus, our main observation of a non-monotonic hysteresis response is supported by this additional calculation. However, the tests do not establish the exact location of the maximum between sampled periods, or guarantee equally reliable periodic steady-state sampling across all temperatures and lattice sizes.

4. Validation Summary and Scope

The results of our validation can be summarized as follows.

Validation test	Main result	Interpretation
Metropolis implementation	444 energy-difference checks and 24 trajectory comparisons agreed	Supports implementation consistency
Initial-state and energy stability	8 of 12 parameter combinations met the selected criteria	Equilibration remains condition-dependent
Exact Boltzmann benchmark	7/8 energy and 4/8 absolute-magnetization comparisons met their precision targets	Supports equilibrium sampling, but some precision tests remain inconclusive
Periodic-state diagnostics	$P = 125$ and 300 supported; $P = 10$ inconclusive	Reliability differs across driving periods
Hysteresis response	Sampled work maximum near $P = 125$ confirmed by neighboring-period comparisons	Supports the principal non-monotonic trend

For statistical comparisons, we calculate uncertainties from independent-run averages rather than treating consecutive Monte Carlo sweeps as independent measurements. The reported intervals are approximate pointwise 95% confidence intervals, and the stability judgments use predefined practical precision targets.

These tests support the basic correctness of our numerical implementation and provide additional evidence for several of our main physical observations.

However, they do not establish convergence for every simulation used in the later sections. In particular, the very short-period simulations remain sensitive to finite observation time and initial conditions, while the broader temperature and finite-size scans have not all undergone the same detailed periodic-state checks.

Therefore, throughout the following analysis, we distinguish the numerical trends directly observed in our simulations from the physical mechanisms proposed to explain them.

The consequences of these remaining uncertainties are discussed in the final Numerical Limitations and Remaining Questions section.

The Driving Period

As we have already introduced, the Hamiltonian of the two-dimensional Ising model is given by:

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i$$

where h is the strength of the external magnetic field.

In this part of the experiment, we are interested in how the system responds to an external magnetic field, and what happens when we change the driving period.

Why does the driving period matter?

We know that different factors can compete with each other in determining the configuration of our spins. As we discussed in the Metropolis algorithm, the energy difference associated with flipping a single spin is:

$$\Delta E = 2s_i \left(J \sum_{j \in \text{nn}(i)} s_j + h \right)$$

where the summation runs over the nearest neighbors of spin i.

The probability of accepting a proposed spin flip is then given by:

$$P_{\text{accept}} = \min\{1, \exp(-\Delta E/T)\}$$

We can see that the energy difference is determined by two contributions: the interaction between a spin and its nearest neighbors, and the external magnetic field. The nearest-neighbor interaction favors the alignment of neighboring spins, while the external field favors spins aligned with the field direction. These two effects may compete when they favor different spin orientations.

When we change the external magnetic field, the system does not necessarily respond immediately. Although the field changes the probability of accepting individual spin flips, the overall magnetization requires many spin updates to evolve. Therefore, the response of the system may lag behind the change in the external field.

Now suppose that we introduce a periodically oscillating magnetic field:

$$h(t) = h_0 \sin\left(\frac{2\pi t}{P}\right)$$

where h_0 is the amplitude of the external field and P is the driving period, measured in Monte Carlo sweeps.

If the driving period is relatively short, the external field may change direction before the magnetization has enough time to respond. On the other hand, with a longer driving period, the system has more time to respond to the changing field and may follow it more closely.

This delayed response can produce a hysteresis loop when we plot the magnetization m against the external field h [4]. Therefore, by changing the driving period, we can investigate how the system's response depends on the timescale of the external driving.

We first demonstrate how the system responds to a change in the external magnetic field:

External Field Response Demonstration

We then introduce a periodically oscillating magnetic field and observe how the magnetization evolves over multiple driving cycles:

Periodic Driving and Hysteresis Demonstration

In the following experiment, we investigate how changing the driving period affects the magnetization dynamics and hysteresis of the system.

To quantitatively investigate the system's behavior, we introduce the magnetization per spin, m, which describes the average spin of the system. Mathematically:

$$m(t) = \frac{1}{L^2} \sum_i s_i(t)$$

The value of m ranges from -1 to 1. A positive m indicates that the system has a net positive magnetization, while a negative m indicates a net negative magnetization. When $m = 1$ or m = -1, all spins are aligned in the same direction.

We then plot m with respect to the external magnetic field h for three different driving periods, $P = 10, 125$, and 300.

We can see that the system shows different responses depending on the driving period. For $P = 125$ and $P = 300$, the trajectories form hysteresis loops, while for $P = 10$, the magnetization remains mostly positive during the observed cycles.

The physical interpretation of a hysteresis loop is that the magnetization does not respond immediately to changes in the external magnetic field. Instead, the response depends on the previous evolution of the system. Therefore, even when the external field has the same value, the magnetization can be different depending on whether the field is increasing or decreasing.

This explains why m is not necessarily a single-valued function of h during periodic driving, and why a hysteresis loop can appear when we plot m against h.

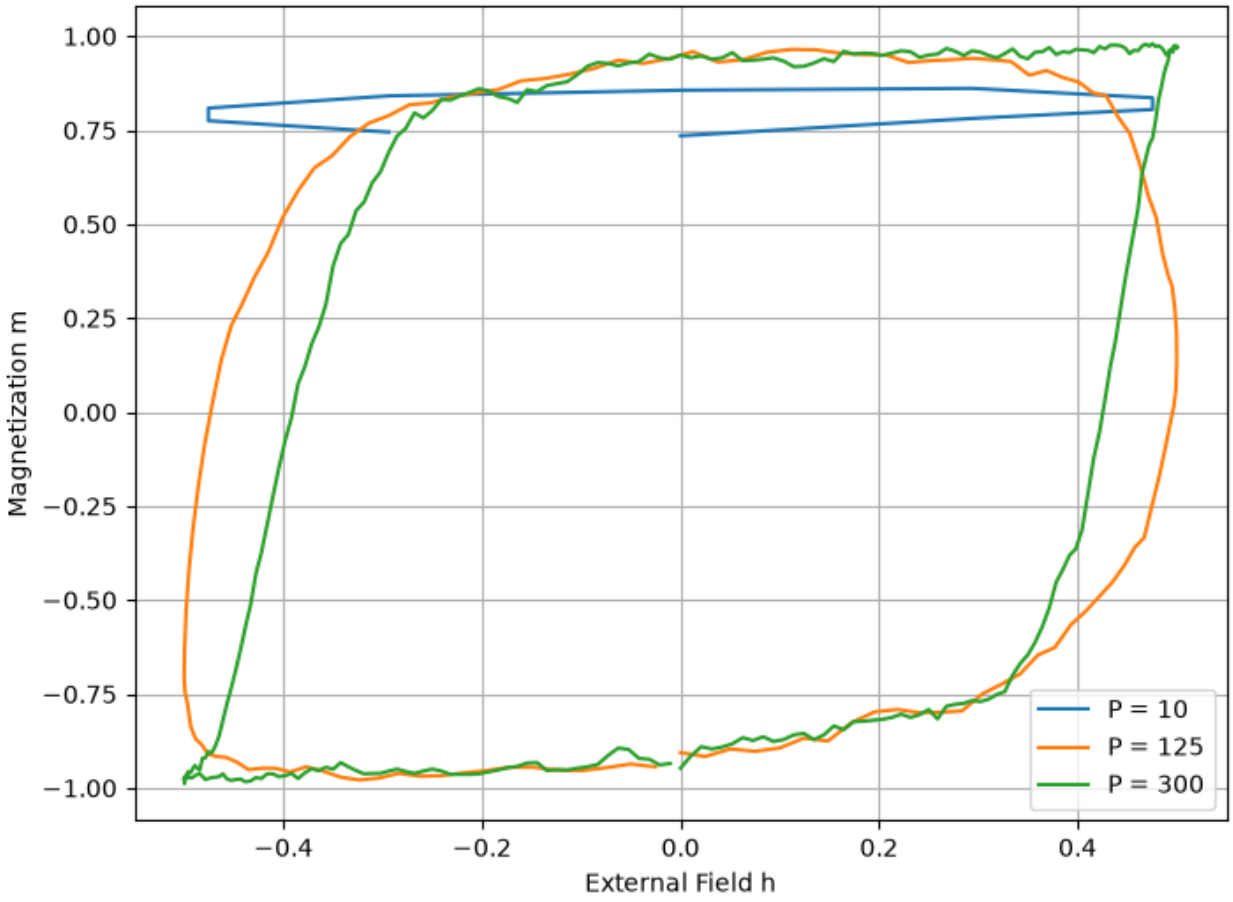


Figure 5. Magnetization versus external magnetic field for driving periods $P = 10$, 125, and 300 MCS. The trajectories illustrate the different hysteretic responses. The corresponding animated demonstrations are available for $P = 10$, $P = 125$, and $P = 300$.

Quantifying the Hysteresis Response

The next step we are interested in is the area of the hysteresis loop for each given driving period P. Since our simulation only provides discrete data points, we need to approximate the loop integral numerically.

As we have already introduced, the Hamiltonian is given by:

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i$$

When we change the external magnetic field h, the work done on the system by this change is given by the partial derivative of H with respect to h, multiplied by dh:

$$\delta W = \frac{\partial H}{\partial h}\, dh$$

Since:

$$\frac{\partial H}{\partial h} = -\sum_i s_i = -Nm$$

where N = L^2 is the total number of spins, the work done per spin is:

$$\delta w = -m\, dh$$

Therefore, over one complete driving cycle, the work done by the external field per spin is:

$$W_{\text{cycle}} = -\oint m\, dh$$

This quantity is related to the signed area enclosed by the hysteresis loop. Once the system reaches a periodic steady state, the average work done over each cycle corresponds to the energy dissipated into the environment as heat.

Since we only have discrete values of magnetization and external field, we can approximate the integral using the trapezoidal rule:

$$W_{\text{cycle}} \approx -\sum_j \frac{m_j + m_{j+1}}{2}(h_{j+1} - h_j)$$

We then investigate how the hysteresis response changes as we increase the driving period P while keeping the other physical parameters fixed.

For the initial driving-period and temperature-dependent investigations, we also use the absolute value of the trapezoidal approximation to the loop integral, denoted by (A), to quantify the magnitude of the hysteretic response.

Although this quantity is closely related to the work performed by the external field, taking the absolute value and using a different numerical integration rule means that it is not strictly identical to the mean signed work.

We therefore distinguish the absolute hysteresis integral (A) from the signed work (W) in the following analyses.

Physical Interpretation

From the figure, we can see that the hysteresis loop area does not change monotonically with the driving period. Instead, it first increases with P, reaches a maximum around $P = 125$ MCS, and then gradually decreases as the period becomes longer.

When P is small, the external magnetic field changes direction very quickly. The system does not have enough time to respond significantly before the field changes again. As a result, the magnetization remains relatively unchanged during a single driving cycle, producing a small hysteresis loop area.

However, as we increase P, the system has more time to respond to the changing external field. The magnetization can undergo larger changes, including reversals between positive and negative values. At the same time, the response still lags behind the external field, producing a larger hysteresis loop.

Interestingly, when P becomes sufficiently large, the loop area begins to decrease again. One possible interpretation is that the system now has more time to follow the external field, so the difference between the magnetization during the increasing and decreasing parts of the field cycle becomes smaller.

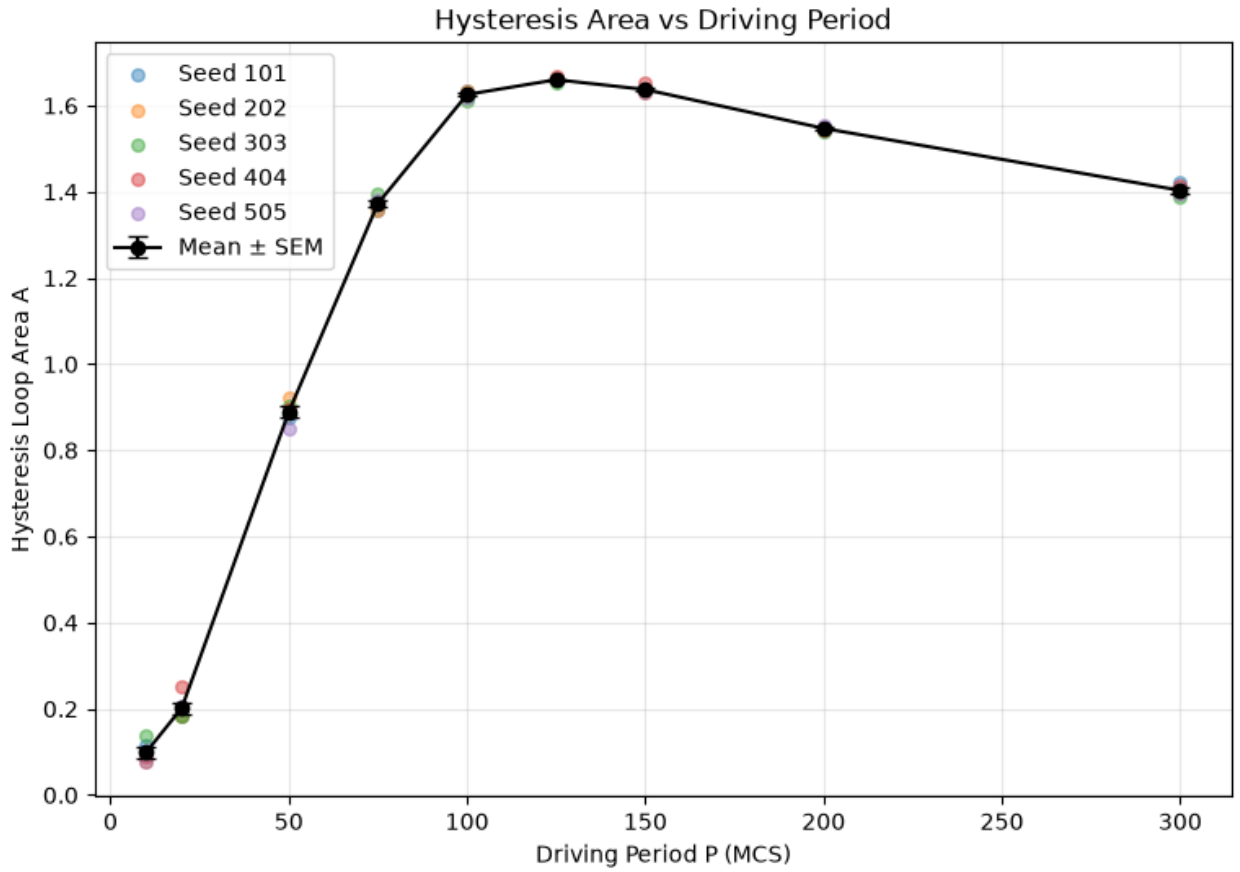


Figure 6. Absolute hysteresis loop integral as a function of driving period, showing individual seeded results and the ensemble mean with standard error.

Therefore, the non-monotonic behavior may be understood as a competition between the timescale of external driving and the characteristic response time of the system.

For very short driving periods, the magnetization cannot respond sufficiently. For intermediate periods, the magnetization responds more strongly but still shows significant hysteresis. For longer periods, the system may follow the external field more closely, reducing the hysteretic response.

This provides a possible physical explanation for the maximum observed at an intermediate driving period. However, further analysis is needed to determine whether the location of this maximum is directly related to a characteristic magnetization reversal time.

Dynamic Order Parameter and Initial-State Dependence

Therefore, from the hysteresis work alone, we cannot always distinguish whether the system barely responds to the external magnetic field or follows the changing field more closely. In both cases, the work done by the external field over one cycle can be small, although the underlying physical behaviors are different.

To further investigate the system's behavior, we introduce another quantitative parameter, the dynamic order parameter Q [4], which describes the average magnetization over one driving period:

$$Q_k = \frac{1}{P} \int_{kP}^{(k+1)P} m(t)\, dt$$

Since our simulation only provides discrete data points, we approximate it by:

$$Q_k \approx \frac{1}{P} \sum_{j=0}^{P-1} m(kP + j)$$

We then calculate the absolute value of Q for each retained cycle and average over multiple cycles and independent runs. This gives us the quantity $\langle |Q| \rangle$.

A large value of |Q| indicates that the system maintains a preferred magnetization direction during the driving cycle. A small value indicates that the positive and negative contributions to the magnetization approximately cancel over the cycle. However, Q alone cannot tell us whether the magnetization actually reverses periodically or simply remains close to zero.

From this figure, we can see that $\langle |Q| \rangle$ decreases significantly as we increase the driving period. For $P = 10$ and $P = 20$, the system maintains a relatively large |Q|, while for longer driving periods, Q gradually approaches zero.

Combined with our previous hysteresis results, this suggests that the small hysteresis response at short and long driving periods may have different physical explanations. Under fast driving, the system may remain magnetized in one direction because the external field changes too quickly for a complete magnetization reversal. For longer periods, the system has more time to respond to the changing field, allowing the positive and negative magnetization contributions to become more balanced.

To further investigate this interpretation, we compare the behavior of the system starting from three different initial configurations: all-up, all-down, and random.

We choose $P = 10, 20$, and 50, keeping the total simulation time fixed at 4000 MCS for each experiment. We also discard the first 2000 MCS before calculating the late-time statistics, and repeat the experiment using eight independent random seeds.

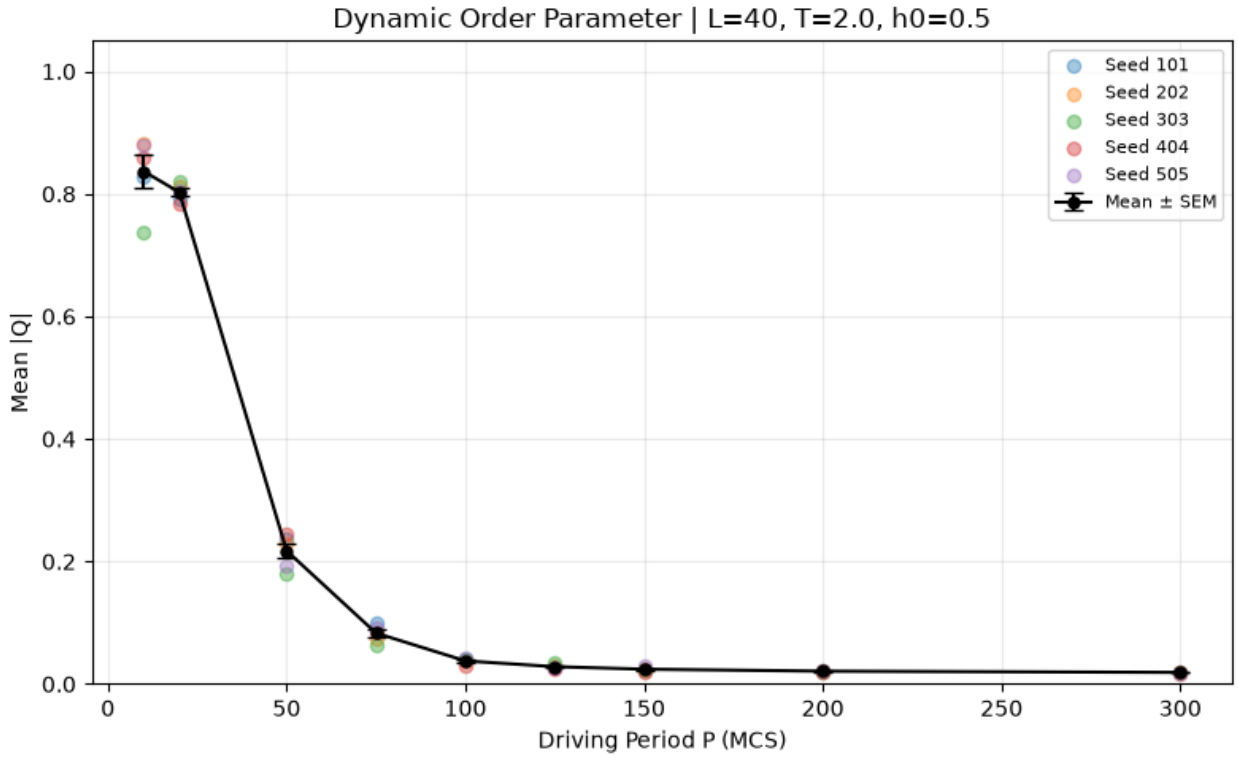


Figure 7. Mean absolute dynamic order parameter $\langle |Q| \rangle$ as a function of driving period for $L = 40$, $T = 2.0$, and $h_0 = 0.5$.

We can see that when P is small, the systems starting from all-up and all-down configurations maintain opposite magnetization directions for a long period of time. For $P = 10$ and $P = 20$, their cycle-averaged magnetizations remain close to +0.8 and -0.8, respectively, throughout most of the simulation.

However, when P increases to 50, the cycle-averaged magnetization fluctuates across zero, and the long-time averages become much less dependent on the initial configuration.

These observations suggest that under fast driving, the external field does not provide enough time for the system to complete a magnetization reversal during each half-cycle. As a result, the system can maintain the magnetization direction established during its earlier evolution, producing a relatively large |Q|.

To further examine whether the initial magnetization determines the long-time behavior, we compare the initial magnetization m_0 with the late-time signed Q for different random initial configurations.

We find that the initial magnetizations are generally close to zero, but the systems can still develop a large positive or negative cycle-averaged magnetization under fast driving.

Interestingly, the sign of the final magnetization does not always agree with the initial magnetization. For example, at $P = 10$, Seed 707 starts with $m_0 = -0.03125$ but eventually reaches $Q_{\text{late}} \approx +0.8743$. Moreover, for Seed 303, the same initial configuration produces $Q_{\text{late}} \approx -0.8684$ at $P = 10$ and $Q_{\text{late}} \approx +0.7999$ at $P = 20$.

These observations suggest that the long-time magnetization direction is not simply determined by the initial net magnetization. Instead, it may depend on the early evolution of the system, during which the ferromagnetic interaction, external driving, and stochastic fluctuations all play a role.

Therefore, the large value of |Q| under fast driving is more appropriately interpreted as a persistent magnetization bias, rather than a simple amplification of the initial magnetization.

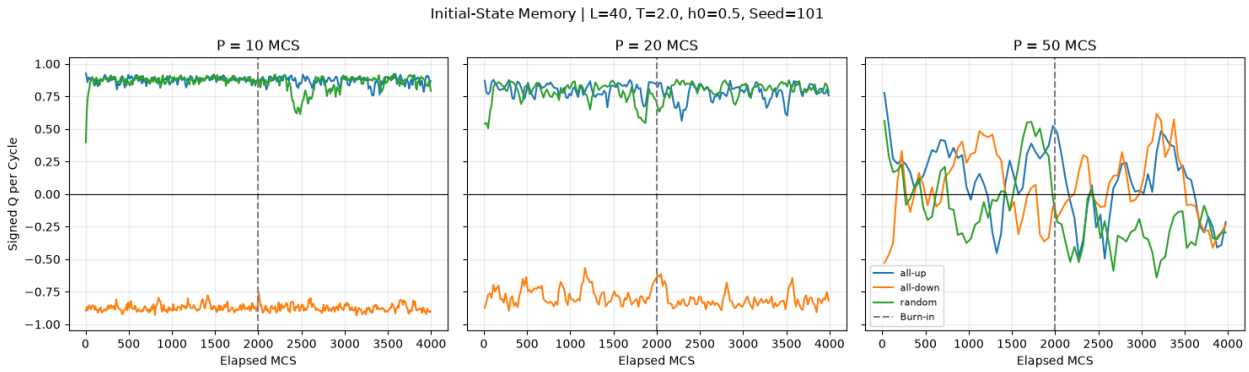


Figure 8. Cycle-averaged signed magnetization versus elapsed Monte Carlo sweeps for all-up, all-down, and random initial configurations at $P = 10$, 20, and 50 MCS.

Another interesting observation is that, when we start from a random configuration, the magnitude of Q can gradually increase over the initial cycles. This suggests that the system may develop a more strongly magnetized configuration during its early evolution.

However, this does not necessarily mean that the oscillating external field directly amplifies the initial magnetization. When $T = 2.0$, which is below the zero-field critical temperature Tc ≈ 2.269, the ferromagnetic interaction already favors the formation of an ordered spin configuration.

In addition, our random initial configurations have magnetizations close to zero, and the final magnetization direction does not always agree with the sign of the initial magnetization. Therefore, the direction of the long-time magnetization may also depend on the early evolution of the system, including the initial external-field phase and stochastic fluctuations.

A possible physical interpretation is that the ferromagnetic interaction promotes the formation of an ordered configuration, while the rapidly oscillating external field is unable to reverse the resulting magnetization within each driving cycle. This allows a preferred magnetization direction to persist over many cycles.

Together, these results explain why the hysteresis work and the dynamic order parameter provide complementary information about the system's response.

Under fast driving, the system may show a small hysteresis response because its magnetization changes relatively little during a cycle. At intermediate periods, the magnetization can undergo larger changes while still lagging behind the external field, producing a larger hysteresis loop. At longer periods, the magnetization may follow the external field more closely, reducing the hysteretic response again.

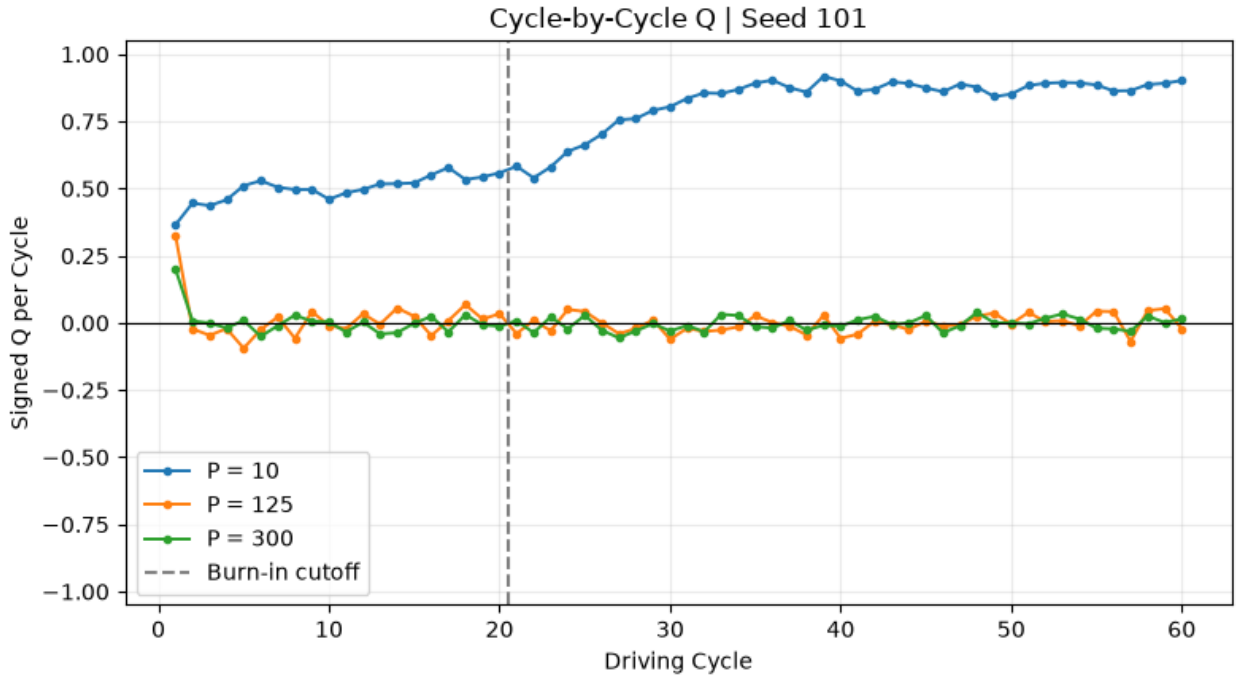


Figure 9. Cycle-by-cycle signed dynamic order parameter for three representative driving periods, with the burn-in cutoff indicated.

Transition to Temperature Dependence

However, there is still an important question: does the persistent magnetization under fast driving mainly result from the short driving period, or does the tendency toward ferromagnetic ordering below Tc also play an important role?

To investigate this question, we next fix the driving period at $P = 10$ and study how the magnetization dynamics change at different temperatures.

By changing T while keeping the other physical parameters fixed, we can examine whether the strong magnetization observed under fast driving persists when thermal fluctuations become more significant.

From this figure, we can see that temperature has a significant effect on the evolution of Q. At lower temperatures, the system gradually develops a strong magnetization and maintains a relatively large |Q|. However, as the temperature increases, the value of |Q| decreases significantly, and the fluctuations become more noticeable near the equilibrium critical temperature.

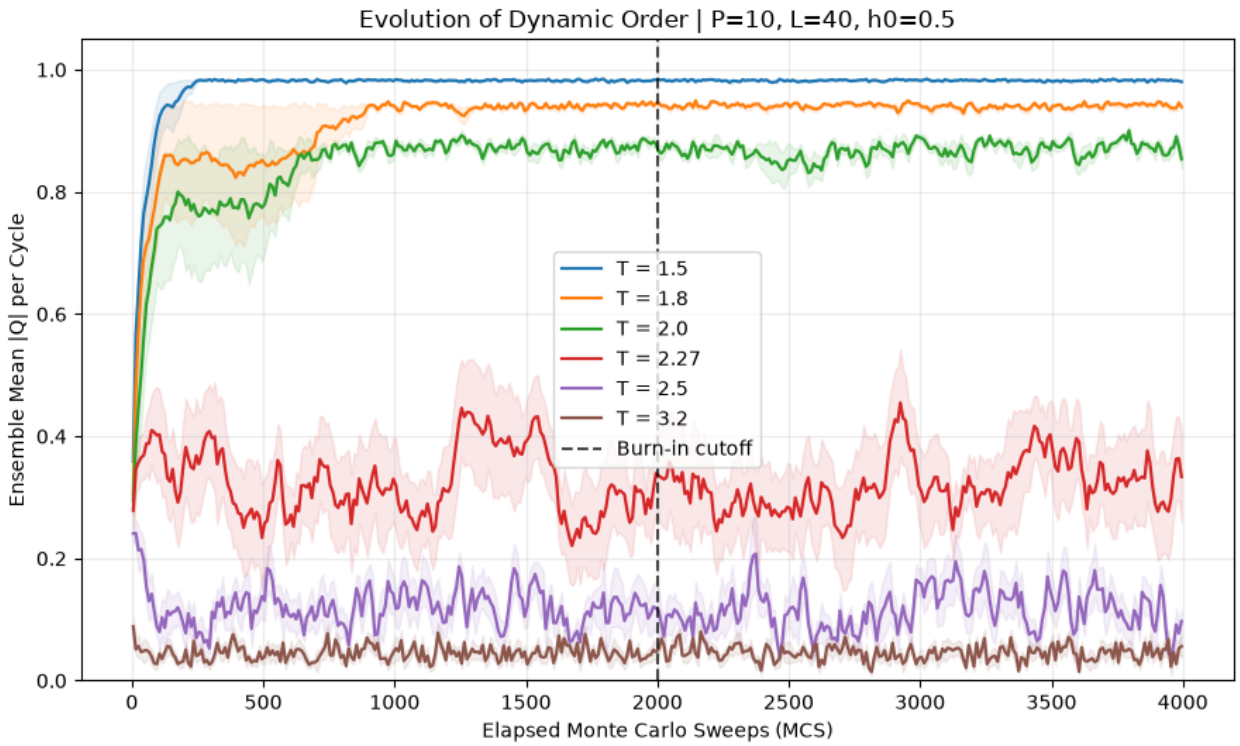


Figure 10. Evolution of the ensemble mean absolute dynamic order parameter under fast driving ($P = 10$) for six temperatures.

To further investigate whether this behavior is mainly determined by the changing external magnetic field or by the intrinsic tendency of the system to form an ordered configuration at a given temperature, we compare the results with and without the periodic external field.

Interestingly, the comparison shows that the effect of the oscillating external field depends strongly on temperature.

At lower temperatures, particularly $T = 1.5$ and $T = 1.8$, the system under fast driving develops a stronger magnetization more quickly than the system without an external field. However, near $T = 2.27$, the opposite behavior is observed: the zero-field system maintains a larger average |Q| than the periodically driven system. At higher temperatures, the difference between the two conditions becomes relatively small.

This suggests that the oscillating external field may affect the ordering dynamics of the system differently at different temperatures, rather than simply increasing or decreasing the magnetization.

One possible explanation is that the external field changes the evolution of magnetic domains, while the ferromagnetic interaction and thermal fluctuations determine how easily an ordered configuration can form and persist. However, our current results are not sufficient to identify the exact mechanism. In particular, the differences observed at lower temperatures may also be influenced by the relatively slow relaxation of the zero-field system.

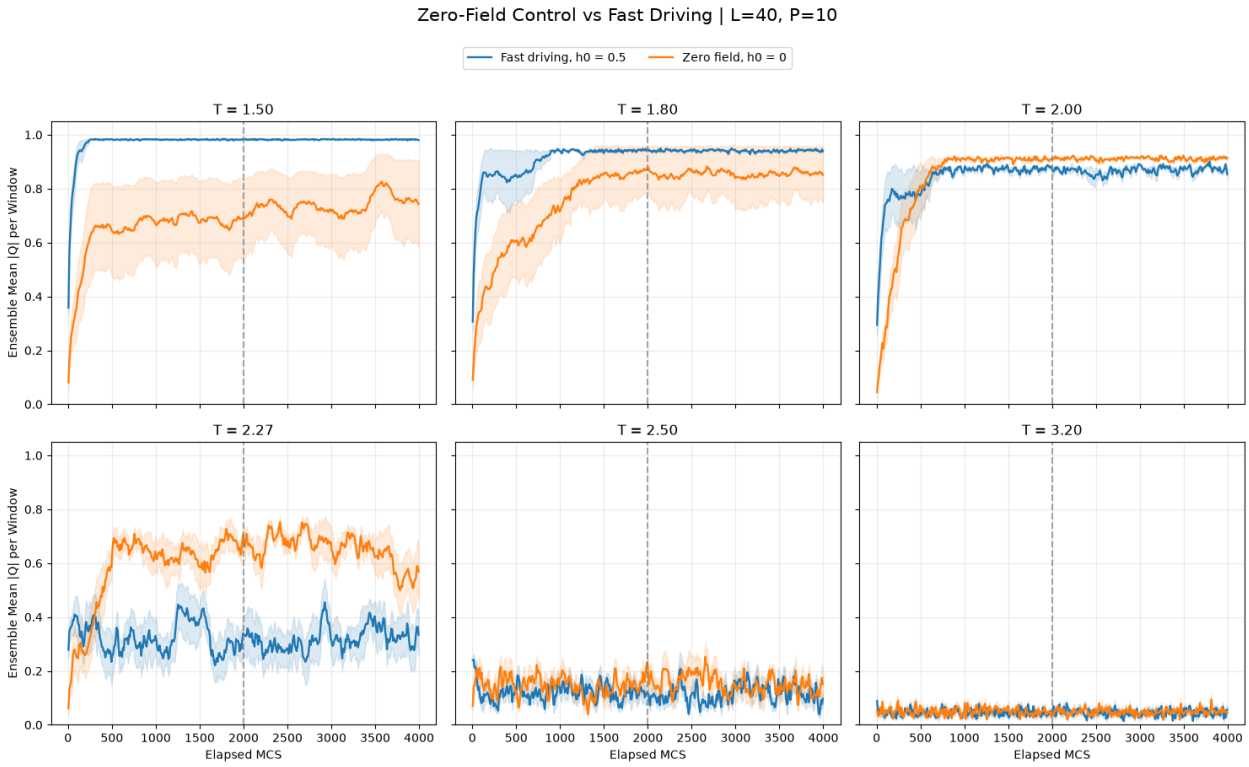


Figure 11. Temperature-dependent comparison between fast periodic driving ($h_0 = 0.5$, $P = 10$) and the zero-field control ($h_0 = 0$).

Therefore, an interesting remaining question is how the periodically oscillating external field affects the formation and evolution of magnetic order at different temperatures, and why it appears to promote stronger magnetization under some conditions but suppress it under others.

Temperature Dependence of the Hysteresis Response

As we have already observed, the behavior of the system can change significantly with temperature, even when the driving period remains the same. A natural question is therefore how the system's response to different driving periods changes when we vary the temperature.

From this figure, we can see that for $T = 1.8, 2.0$, and 2.3, the hysteresis loop area shows a similar non-monotonic trend: it first increases with the driving period and then decreases after reaching a maximum.

One possible physical interpretation is that when the driving period is short, the external magnetic field changes too quickly for the system to fully respond. As a result, the magnetization does not change significantly during each cycle, producing a relatively small hysteresis loop.

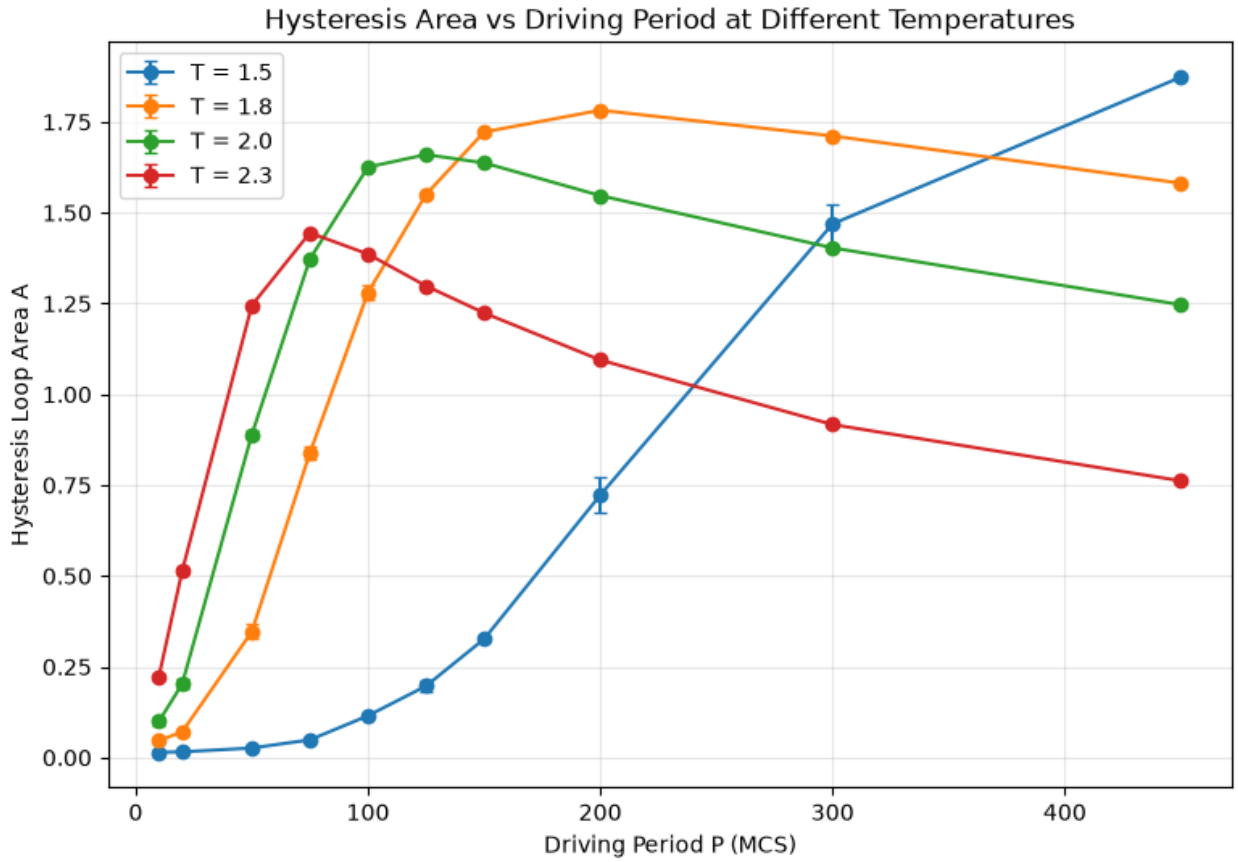


Figure 12. Absolute hysteresis loop integral versus driving period at four different temperatures.

As we increase the driving period, the system has more time to respond to the changing field, allowing larger changes in magnetization. However, the magnetization still lags behind the external field, resulting in a larger hysteresis loop.

When the driving period becomes sufficiently long, the system may follow the external field more closely, reducing the difference between the magnetization during the increasing and decreasing parts of the field cycle. Therefore, the hysteresis loop area begins to decrease again.

Interestingly, we also observe that as the temperature increases, the driving period corresponding to the maximum hysteresis response becomes shorter. One possible explanation is that stronger thermal fluctuations make it easier for the system to escape from its existing magnetized configuration and develop a magnetization in the opposite direction. Therefore, the system may require less time to respond to the changing external field at higher temperatures.

However, this interpretation is still based on the observed hysteresis response, rather than a direct measurement of the magnetization reversal time.

For $T = 1.5$, the hysteresis area continues to increase within our original range of driving periods. Therefore, we extend the period scan to $P = 450, 600$, 800, and 1000 MCS.

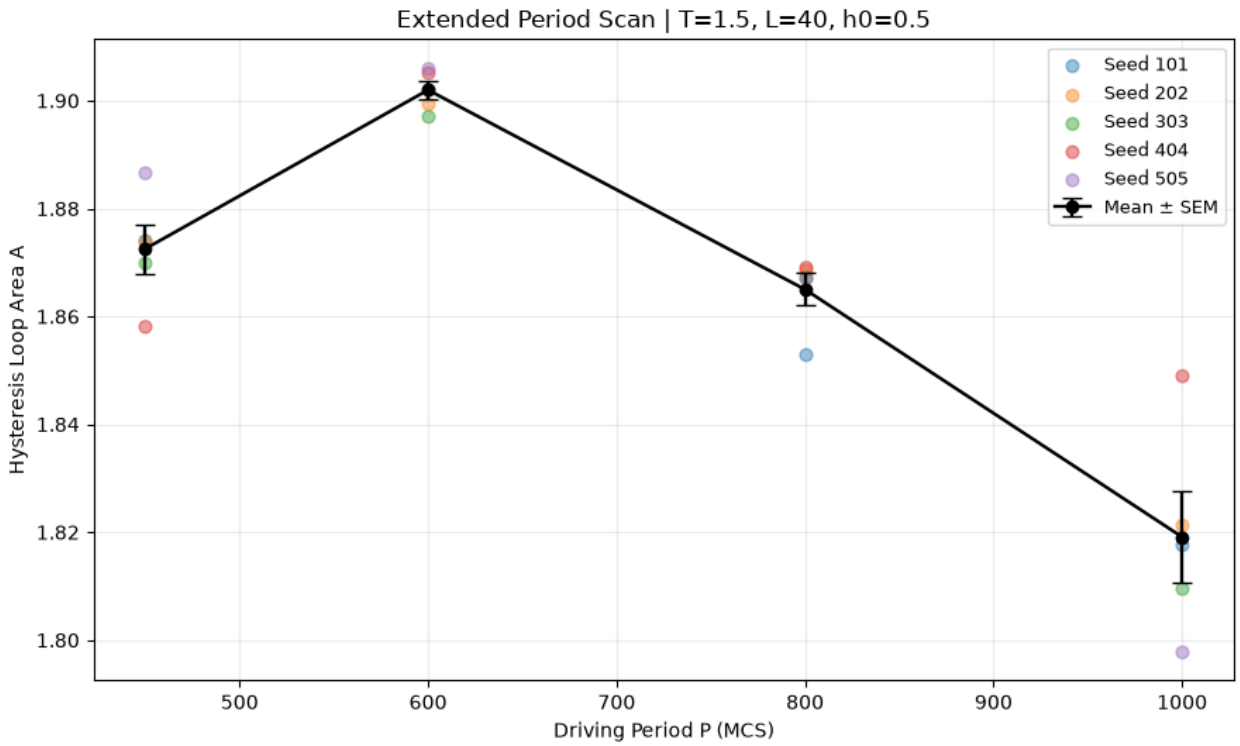


Figure 13. Extended driving-period scan of the absolute hysteresis loop integral at $T = 1.5$, including seeded results and ensemble means.

From this additional experiment, we can see that the hysteresis area also shows a non-monotonic trend at $T = 1.5$. Among the investigated periods, the maximum appears around $P = 600$ MCS, after which the loop area gradually decreases.

This observation is consistent with the trend found at higher temperatures, but with the maximum response occurring at a much longer driving period.

A possible physical explanation is that at lower temperatures, thermal fluctuations become weaker, making the existing magnetized configuration more stable. Consequently, the system may require a longer time to complete a magnetization reversal when the external field changes direction.

Together, these results suggest that the characteristic driving period associated with the maximum hysteresis response increases as the temperature decreases. This is consistent with our interpretation that the hysteresis response is influenced by the competition between the external driving timescale and the magnetization reversal dynamics of the system.

To further investigate how temperature affects the magnetization dynamics, we calculate the dynamic order parameter for different driving periods and temperatures.

From the figure, we can see that the dynamic order parameter generally decreases as the driving period increases. However, the period at which this decrease occurs strongly depends on temperature.

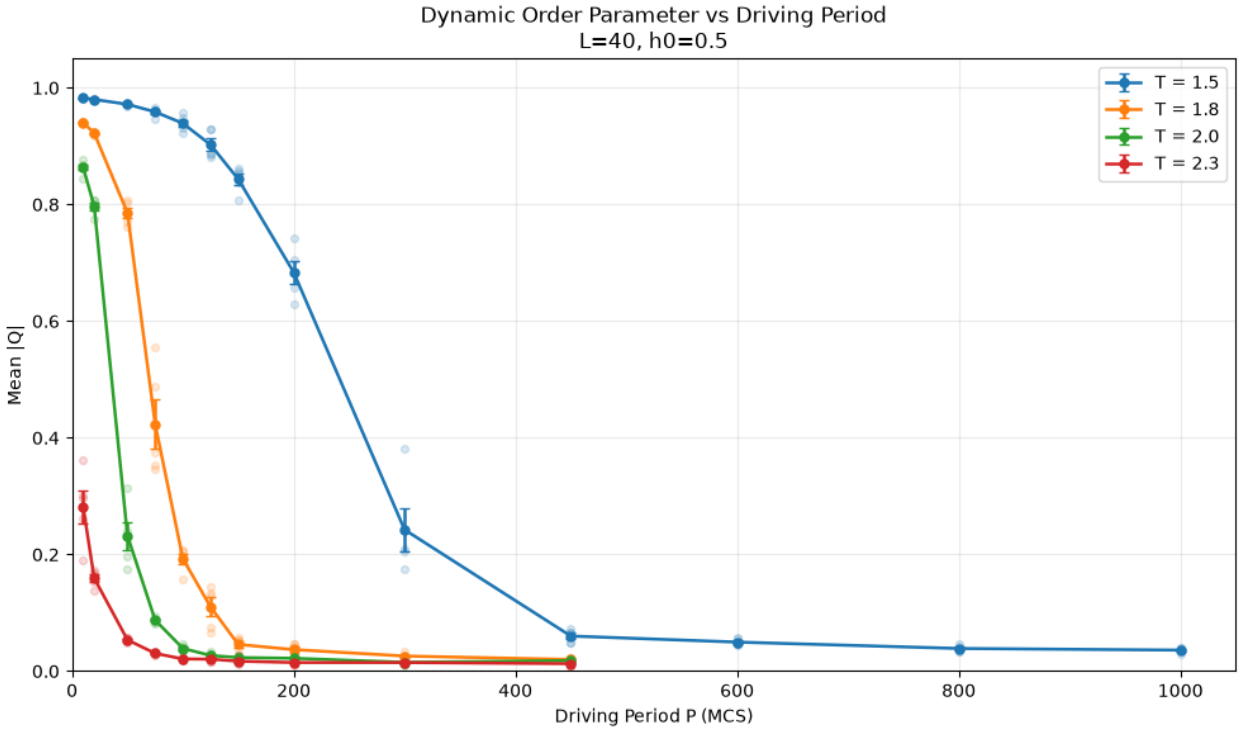


Figure 14. Mean absolute dynamic order parameter versus driving period at different temperatures for $L = 40$ and $h_0 = 0.5$.

At lower temperatures, particularly $T = 1.5$, the system maintains a large |Q| over a much wider range of driving periods. As the temperature increases, the decrease in |Q| occurs at shorter periods.

This is consistent with our previous observation that the driving period corresponding to the maximum hysteresis response also becomes shorter as temperature increases.

Interestingly, the decrease in |Q| does not necessarily occur at the same period as the maximum hysteresis area. For example, at $T = 1.5$, the dynamic order parameter has already decreased significantly before the hysteresis area reaches its observed maximum near $P = 600$ MCS.

One possible interpretation is that the system first becomes able to reverse its magnetization more symmetrically as the driving period increases, while the magnetization response can still significantly lag behind the external field.

Therefore, even when the period-averaged magnetization approaches zero, the hysteresis loop can remain relatively large.

Together, these observations suggest that temperature affects the competition between the driving timescale and the magnetization response of the system. At lower temperatures, the system may require a longer driving period to reverse its magnetization and develop a more symmetric periodic response.

However, we have not directly measured the characteristic magnetization reversal time, so this remains a possible physical explanation rather than a quantitatively established relationship.

Supplementary visualization: Multi-Temperature Period Explorer provides an animated comparison of the magnetic response across different temperatures and driving periods.

Finite-Size Dependence

Lastly, we want to investigate whether our previous conclusions can be generalized to different lattice sizes, and to what extent a finite lattice can approximate the behavior of the two-dimensional Ising model in the thermodynamic limit [5].

To investigate this, we calculate the mean signed work per spin and the mean absolute value of the dynamic order parameter Q for different lattice sizes, $L = 20, 40$, and 80, while keeping $T = 2.0$ and h_0 = 0.5 fixed.

From this figure, we can see that the mean signed work shows a relatively similar dependence on the driving period for different lattice sizes. All three curves reach their maximum around $P = 125$ MCS, and their overall shapes are quite consistent. However, there are still some differences between the lattice sizes, particularly at shorter and intermediate driving periods.

Interestingly, the dynamic order parameter shows a much stronger finite-size dependence. We observe that as the lattice size increases, the value of $\langle |Q| \rangle$ decreases significantly, especially in the region where Q changes rapidly with the driving period.

For example, at $P = 50$, the mean absolute dynamic order parameter decreases substantially from $L = 20$ to $L = 80$. This raises an important question: is this difference a genuine finite-size effect, or could it be caused by insufficient equilibration and the limited measurement time in our original simulations?

To investigate this, we perform an additional numerical test at $P = 50$, where the observed finite-size difference is particularly significant.

In this test, we increase the burn-in period to 3000 MCS and the measurement period to 6000 MCS. We also compare three different initial configurations: all-up, all-down, and random, using eight independent seeds for each configuration.

In addition, we divide the measurement period into two equal windows and compare the mean observables obtained from each half. This allows us to check whether the measured values are still undergoing systematic changes during the observation window.

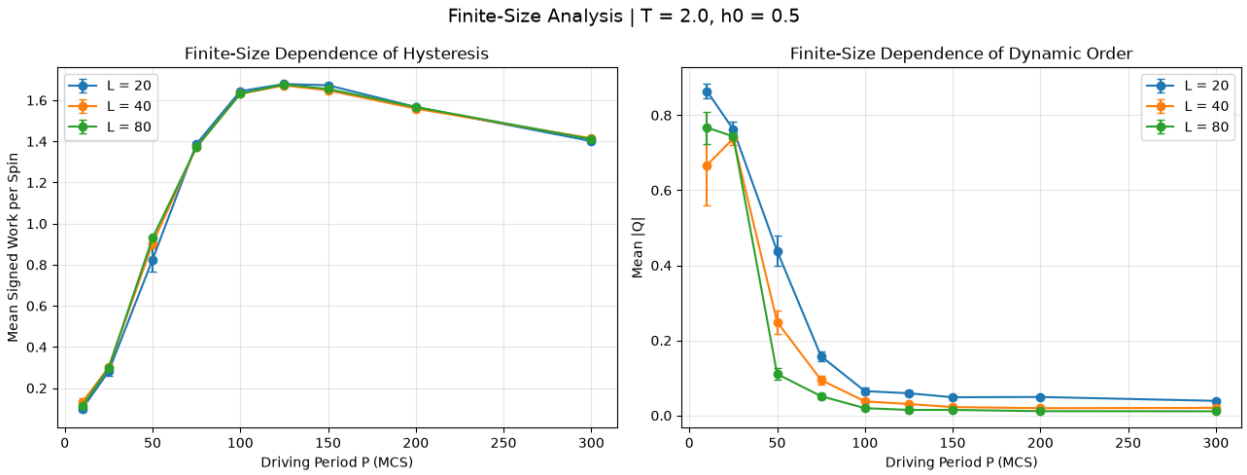


Figure F1. Finite-size dependence of the mean signed work per spin (left) and mean absolute dynamic order parameter (right) for $L = 20, 40$, and 80, at $T = 2.0$ and h_0 = 0.5. The results are obtained using 25 driving cycles, with the first 10 cycles discarded, and three independent seeds per parameter combination. Error bars represent the standard error of the mean across seeds.

From this additional experiment, we observe that the strong finite-size dependence of Q remains even after increasing the simulation time and using different initial configurations.

At $L = 20$, the mean absolute dynamic order parameter remains around 0.39–0.40, while at $L = 40$ it decreases to approximately 0.20–0.23. For $L = 80$, it further decreases to around 0.10–0.11.

The results from different initial configurations are also generally consistent within their statistical uncertainties. This suggests that the observed decrease in Q is unlikely to be explained entirely by the choice of initial configuration.

Furthermore, the differences between the first and second measurement windows are statistically consistent with zero within the reported uncertainties. This provides additional evidence that the measured values are relatively stable over our selected observation time, although it does not guarantee convergence over arbitrarily long times.

The signed work, on the other hand, shows a smaller relative change with lattice size. We observe an increase from approximately 0.86–0.87 at $L = 20$ to around 0.92 at $L = 40$ and $L = 80$.

Therefore, the hysteresis work is not completely independent of lattice size, but its size dependence is considerably weaker than that of the dynamic order parameter at $P = 50$.

To further examine the magnetization dynamics, we also inspect the signed Q for individual driving cycles.

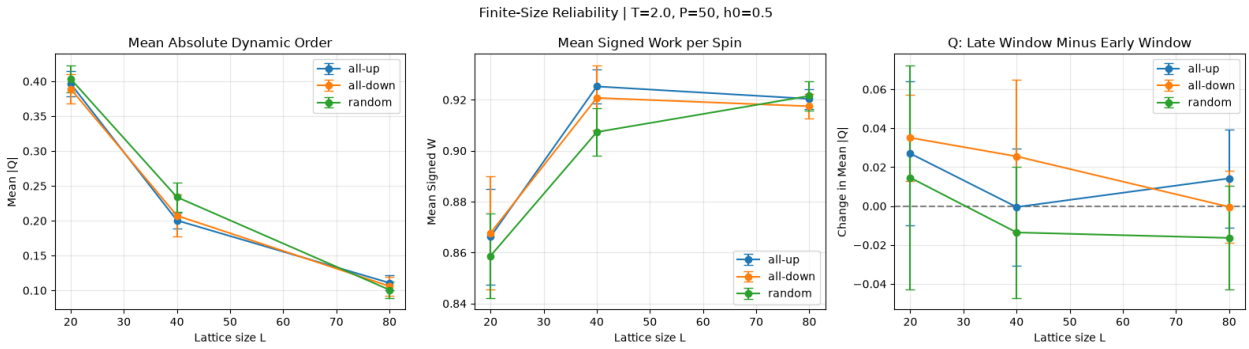


Figure F2. Additional finite-size reliability test at $P = 50$ MCS, $T = 2.0$, and h_0 = 0.5. The panels show the mean absolute dynamic order parameter, mean signed work per spin, and the difference in mean absolute dynamic order between the second and first halves of the measurement window. Each initial-state group contains eight independent runs, with 3000 MCS discarded as burn-in and 6000 MCS retained for measurement. Error bars indicate approximate 95% confidence intervals calculated across independent runs.

For $L = 80$, the representative trajectories from different initial configurations fluctuate around zero rather than maintaining a persistent magnetization in one direction throughout the measurement window.

This is consistent with the relatively small value of $\langle |Q| \rangle$ observed for larger lattices.

One possible physical explanation for the difference between W and Q is that the two quantities measure different aspects of the system's response. The hysteresis work depends on how the magnetization changes relative to the external field, while Q measures the average magnetization over a complete driving cycle.

Since we take the absolute value of Q before averaging, finite-size fluctuations can contribute to a positive $\langle |Q| \rangle$ even when there is no persistent preference for either magnetization direction. In addition, the magnetization reversal dynamics may themselves depend on lattice size.

Our results do not yet distinguish these two possible contributions.

Overall, the additional test provides stronger evidence that the finite-size dependence of the dynamic order parameter is a reproducible feature of our simulations at $P = 50$, rather than simply a consequence of the original short measurement window.

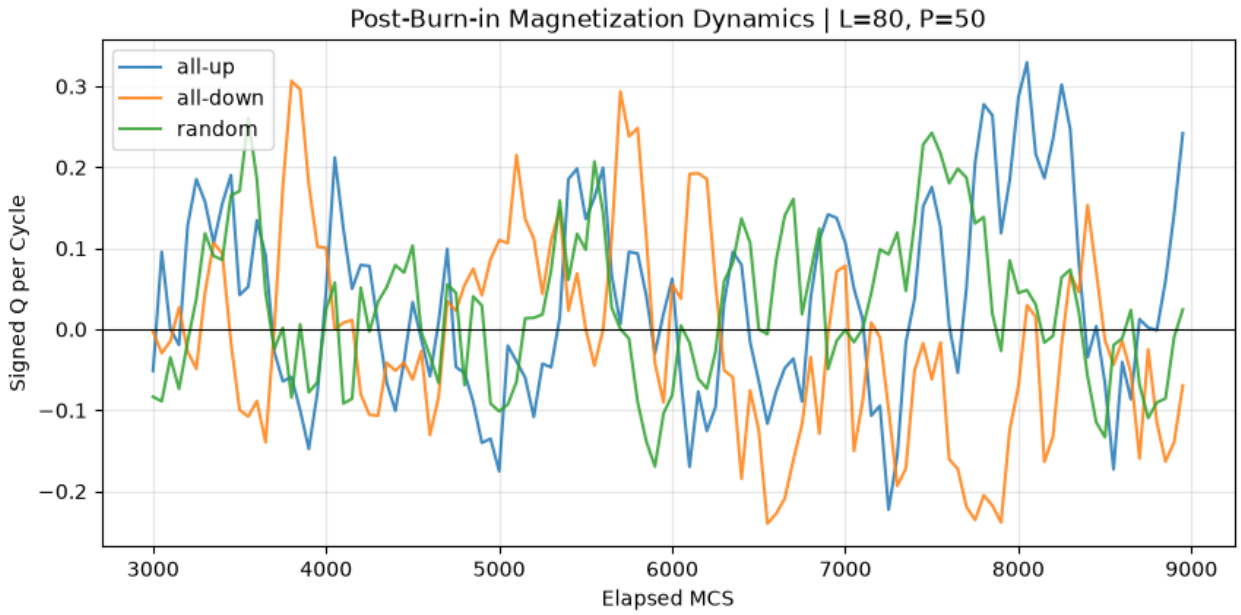


Figure F3. Representative post-burn-in cycle-averaged signed magnetization trajectories for $L = 80$, $P = 50$, $T = 2.0$, and $h_0 = 0.5$. Each curve corresponds to one seeded run starting from an all-up, all-down, or random initial configuration. The trajectories illustrate fluctuations in the signed dynamic order parameter during the measurement window and are not intended to represent the full ensemble statistics.

However, since we investigate only three finite lattice sizes and a limited range of simulation times, we cannot directly establish the behavior in the thermodynamic limit.

A more systematic investigation using larger lattices, longer simulations, and additional characteristic driving periods would be required to determine how the two observables approach their infinite-system behavior.

Limitations and Scope of Conclusions

Overall, from our numerical validation, we can see that although our implementation of the Metropolis algorithm is supported by several independent tests, the numerical results do not satisfy all the stability criteria under certain combinations of physical parameters. Therefore, some of our observations are more reliable than others, and the physical conclusions we can draw from them should be considered with these limitations in mind.

1. Driving Period Dependence

For the driving period investigation, we observed a non-monotonic dependence of the hysteresis response on P, with the maximum occurring around $P = 125$ MCS for $T = 2.0$ and $L = 40$.

This observation is supported by our additional numerical validation, where the signed work also reaches its largest value near the same driving period, and the results at $P = 125$ and $P = 300$ satisfy our selected stability criteria.

However, the behavior at very short driving periods remains less certain. In particular, for $P = 10$, the dynamic order parameter shows sensitivity to the initial configuration, burn-in period, and measurement duration.

Therefore, although we observe persistent magnetization over many driving cycles, we cannot conclude that this behavior represents a fully sampled periodic steady state or that the magnetization direction will remain unchanged indefinitely.

Our interpretation that fast driving prevents the system from completing magnetization reversal is consistent with the observed behavior, but a more detailed investigation of the long-time dynamics would be required to establish this mechanism quantitatively.

2. Temperature Dependence

In the temperature investigation, we found that the driving period corresponding to the maximum hysteresis response generally becomes longer as the temperature decreases. The dynamic order parameter also shows a similar temperature-dependent shift in the region where its value decreases significantly.

These observations suggest that the response of the system depends on the competition between thermal fluctuations and the external driving timescale.

However, our additional periodic-state validation was performed only for selected conditions at $T = 2.0$. We have not established the same level of numerical convergence for every temperature and driving period investigated.

Furthermore, although we proposed that the observed shift is related to the magnetization reversal time, we have not directly measured this characteristic timescale. Therefore, this interpretation remains a possible physical explanation rather than a quantitatively verified relationship.

Our zero-field comparison also raises an additional question. At lower temperatures, fast driving appears to accelerate the development of strong magnetization, while near the equilibrium critical temperature, it can produce a smaller cycle-averaged magnetization than the zero-field system.

The underlying mechanism remains unclear, and further investigation of magnetic domain evolution would be needed to explain this temperature-dependent behavior.

3. Finite-Size Dependence

Finally, our finite-size investigation shows that the mean signed work per spin remains relatively consistent across $L = 20, 40$, and 80, while the dynamic order parameter shows more noticeable differences, particularly at intermediate driving periods.

This suggests that the sensitivity to lattice size depends on the observable we are investigating.

However, our simulations include only three lattice sizes, and we have not established equally strong convergence for every size and driving period. In addition, the magnetization reversal dynamics may themselves depend on lattice size.

Therefore, we cannot determine whether the observed changes in Q are mainly caused by finite-size fluctuations or by changes in the magnetization reversal dynamics.

More importantly, the agreement between the hysteresis curves for three finite lattice sizes does not prove that the results have reached the thermodynamic limit.

A more systematic investigation using larger lattices and longer simulations would be required to determine how these observables behave as L approaches infinity.

Overall Assessment

Despite these limitations, the numerical results consistently show several important qualitative trends: a non-monotonic hysteresis response to the driving period, a temperature-dependent shift in the observed maximum response, and different finite-size sensitivities for the hysteresis work and the dynamic order parameter.

Our validation provides additional support for the numerical implementation and selected results, but it does not establish the same level of reliability across the entire parameter space.

Therefore, the main conclusions of this study should be understood as numerical observations over finite lattice sizes and simulation times, rather than rigorous statements about the infinite-system limit or the existence of a dynamic phase transition.

Conclusion

In this study, we investigated the nonequilibrium behavior of the two-dimensional Ising model under a periodically oscillating external magnetic field, focusing on how the system's magnetization response changes with the driving period, temperature, and lattice size.

We first found that the hysteresis response shows a non-monotonic dependence on the driving period. When the period is short, the system cannot fully respond to the rapidly changing external field, resulting in a relatively small hysteresis loop. As the period increases, the magnetization can respond more strongly while still lagging behind the external field, producing a larger hysteresis response. At sufficiently long periods, the response becomes less hysteretic, and the loop area begins to decrease.

By introducing the dynamic order parameter Q, we also investigated whether the system maintains a preferred magnetization direction over each driving cycle. We observed that strong cycle-averaged magnetization can persist under fast driving, while increasing the driving period generally reduces the magnitude of Q.

We then extended our investigation to different temperatures. An important observation is that the driving period corresponding to the maximum hysteresis response becomes longer as the temperature decreases. The dynamic order parameter shows a related temperature dependence, with its main decrease also shifting toward longer periods at lower temperatures. These results are consistent with the idea that the system's response is determined by the competition between thermal fluctuations, magnetization reversal dynamics, and the timescale of the external driving.

Finally, we investigated the finite-size dependence of our results. We found that the mean signed work per spin remains relatively consistent across the investigated lattice sizes, while the dynamic order parameter shows a stronger dependence on lattice size, particularly at intermediate driving periods. This suggests that different observables can have different sensitivities to the finite size of the numerical system.

Our numerical validation supports the basic implementation of the Metropolis algorithm and several of the main qualitative observations. However, some parameter combinations, especially under very fast driving, remain sensitive to the simulation duration and initial conditions. Therefore, our results should be understood within the investigated parameter ranges and observation times, rather than as a rigorous determination of the thermodynamic limit or a dynamic phase transition.

Overall, this project demonstrates how a relatively simple numerical model can reveal complex nonequilibrium behavior. More importantly, by comparing different physical conditions and examining the reliability of our numerical results, we developed a clearer understanding of how magnetic ordering, thermal fluctuations, and external driving interact in determining the system's dynamics.

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